Digital Stethoscope using ESP32 and INMP441

Project Report

Abstract

The **Digital Stethoscope** is a smart biomedical device designed to monitor and visualize heart and lung sounds using an ESP32 microcontroller and an INMP441 digital microphone sensor. The system captures body sounds, processes them using digital filtering techniques, and displays waveforms and BPM information on an OLED display.

The project provides two operating modes:

- Heart Sound Monitoring
- Lung Sound Monitoring

The device offers a portable and low-cost solution for biomedical signal monitoring and educational healthcare applications.

Introduction

Traditional stethoscopes rely on acoustic amplification and manual interpretation of sounds by medical professionals. In contrast, digital stethoscopes provide better sound processing, visualization, and monitoring capabilities.

This project implements a digital stethoscope capable of:

- Capturing body sounds
- Filtering unwanted noise
- Displaying waveforms on an OLED
- Detecting heartbeats and BPM
- Monitoring breathing sounds

The system uses digital signal processing techniques to improve the quality of heart and lung sound analysis.

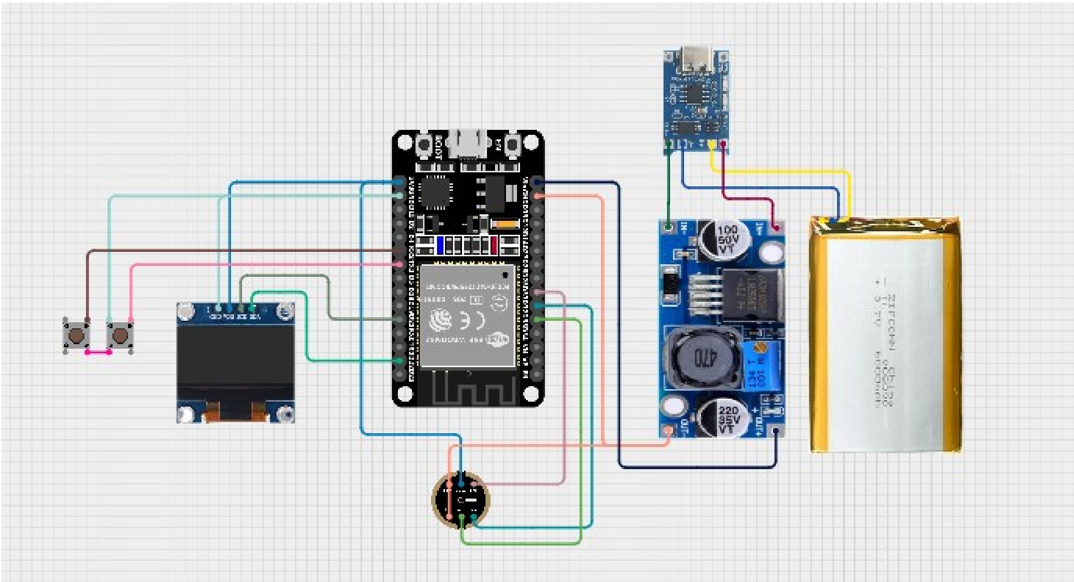
Objectives

The main objectives of this project are:

- To develop a digital stethoscope using ESP32.
- To acquire heart and lung sounds using the INMP441 microphone.
- To filter and process biomedical audio signals.
- To display waveforms on an OLED screen.
- To calculate and display heart BPM.
- To create a simple button-controlled user interface.

Components Used

Component	Function
ESP32	Main controller
INMP441 Microphone	Audio signal acquisition
OLED Display SSD1306	Display waveforms and BPM
Push Buttons	Mode selection and reset
Jumper Wires	Circuit connections



Working Principle

Audio Acquisition

The INMP441 digital microphone captures body sounds such as heartbeat and breathing signals. The ESP32 receives the audio data using the I2S communication interface.

The audio is sampled at:

- **8 kHz sampling rate**

Signal Processing

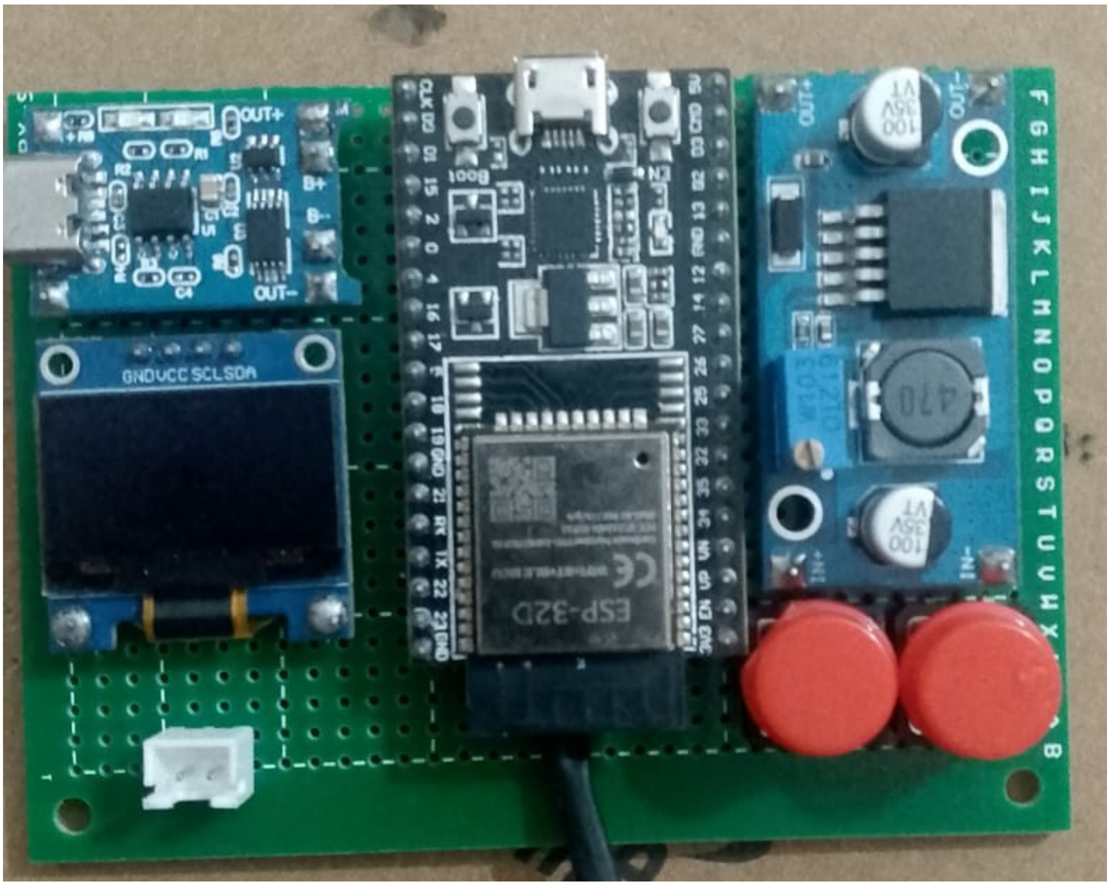
The captured audio signals are processed using digital filters.

Heart Mode

A bandpass filter is applied between:

- **30 Hz – 300 Hz**

This range isolates heart sounds and reduces unwanted noise.



Lung Mode

A bandpass filter is applied between:

- **150 Hz – 1000 Hz**

This range captures breathing and lung-related sounds.

BPM Detection

In Heart Mode, the system detects peaks in the audio envelope to calculate:

- Beats Per Minute (BPM)

The BPM value is continuously updated and displayed on the OLED screen.

OLED Display Interface

The OLED display provides:

- Welcome screen
- Heart mode display
- Lung mode display
- Real-time waveform visualization

- BPM display

Heart Mode

- Heart icon
- BPM value
- Heart sound waveform

Lung Mode

- Lung icon
- Breath monitoring waveform

User Controls

Two push buttons are used for system control:

Button	Function
Mode Button	Switch between modes
Reset Button	Reset BPM counter

The interface is simple and user-friendly.

Software Description

The system is programmed in **Arduino C/C++** using the following libraries:

Library	Purpose
driver/i2s.h	Audio data acquisition
Adafruit_GFX.h	Graphics handling
Adafruit_SSD1306.h	OLED control
Wire.h	I2C communication
math.h	Signal processing calculations

Features

- Digital heart sound monitoring
- Digital lung sound monitoring
- Real-time waveform display
- BPM calculation
- OLED graphical interface
- Portable embedded design
- Noise reduction using filters

Advantages

- Low-cost biomedical device
- Portable and lightweight

- Easy to operate
- Real-time visualization
- Improved sound monitoring
- Educational healthcare application

Limitations

- Sensitive to external noise
- Requires proper sensor placement
- Not intended for clinical diagnosis
- Basic BPM detection accuracy

Applications

- Biomedical engineering projects
- Medical training systems
- Digital healthcare prototypes
- Signal processing education
- Smart diagnostic research

Future Improvements

Future enhancements may include:

- Wireless audio transmission
- Mobile application connectivity
- Audio recording and storage
- AI-based abnormality detection
- Noise cancellation improvement
- Battery-powered portable design

Conclusion

The **Digital Stethoscope using ESP32 and INMP441** successfully demonstrates biomedical sound acquisition and digital signal processing techniques. The system can monitor heart and lung sounds, display waveforms in real time, and calculate BPM using embedded hardware and software.

This project highlights the importance of embedded systems and digital healthcare technologies in modern biomedical applications.